

BEMIDJI, MN, USAWMO: **727550**Lat: **47.500N**Lon: **94.933W**Elev: **1392**StdP: **13.97**Time Zone: **-6.00 (NAC)**Period: **94-19**WBAN: **14958****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.1	-17.5	-28.0	1.2	-18.3	-25.1	1.4	-14.0	26.1	13.7	23.5	13.8	5.3	260	0.645

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	86.0	69.7	82.4	67.4	80.7	66.0	72.6	82.3	70.3	79.6	68.4	77.1	12.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.7	115.0	78.9	67.0	104.5	75.3	64.8	96.6	73.6	37.2	82.1	35.1	79.1	33.5	77.3	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.9	20.5	18.5	-30.6	91.5	5.8	3.1	-34.8	93.7	-38.2	95.5	-41.5	97.2	-45.7	99.5		
(5)			-28.9	76.3	5.2	2.2	-32.7	77.9	-35.7	79.3	-38.6	80.5	-42.3	82.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.8	7.8	12.0	25.5	39.6	52.5	62.7	67.6	65.2	56.8	43.5	27.9	14.2
	DBStd	23.31	14.47	13.79	13.15	10.17	8.66	6.89	5.82	6.06	8.46	9.04	11.48	13.21
	HDD50	5591	1307	1065	762	335	74	3	0	0	29	243	664	1109
	HDD65	9494	1772	1485	1225	763	398	119	40	72	267	667	1112	1574
	CDD50	1856	0	0	2	22	150	385	546	472	234	43	2	0
	CDD65	284	0	0	0	0	9	52	121	79	22	1	0	0
	CDH74	2713	0	0	0	11	149	504	1098	706	224	21	0	0
	CDH80	642	0	0	0	2	30	119	281	163	44	3	0	0
	WSAvg	8.4	8.2	8.3	8.6	9.1	9.1	8.3	7.7	7.5	8.3	9.1	8.8	8.2
	PrecAvg	26.9	0.7	0.6	1.1	1.8	3.0	4.1	4.2	3.1	3.2	2.9	1.3	0.9
(15) Precipitation	PrecMax	34.8	2.2	1.2	2.8	3.9	6.3	7.2	7.8	6.1	7.4	5.1	3.7	1.8
	PrecMin	19.9	0.1	0.1	0.3	0.6	1.1	0.9	1.6	1.1	1.3	0.2	0.0	0.2
	PrecStd	3.5	0.5	0.3	0.6	0.9	1.3	1.6	1.7	1.5	1.4	1.4	0.9	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.0	45.6	63.0	73.3	83.9	89.1	90.3	89.7	84.4	76.9	61.2	43.4
		MCWB	35.7	38.3	50.6	55.3	64.0	71.2	73.3	72.2	68.5	58.4	50.8	36.2
	2%	DB	34.6	39.4	54.1	66.5	78.7	83.2	86.1	84.0	80.7	69.0	54.1	37.5
		MCWB	31.7	35.3	45.3	50.5	60.1	67.6	70.6	69.0	65.9	56.4	46.6	33.6
	5%	DB	31.6	36.2	47.5	63.0	73.0	79.4	82.4	81.1	75.6	63.2	47.9	34.4
		MCWB	30.1	32.7	40.6	48.0	56.9	64.2	68.4	66.4	63.4	53.2	42.1	31.7
	10%	DB	27.9	32.3	42.9	57.1	68.8	76.5	80.8	78.5	72.1	57.4	43.5	31.5
		MCWB	25.9	29.9	37.1	45.2	55.1	62.8	67.1	65.2	61.5	49.5	38.3	30.1
	0.4%	WB	36.1	39.6	53.1	57.3	67.5	73.5	75.9	74.4	71.7	62.3	52.9	38.1
		MCDB	41.2	44.2	57.8	70.7	78.3	84.9	85.5	85.0	80.8	71.5	57.3	41.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.4	36.0	45.8	52.2	62.9	70.1	73.3	71.4	68.4	58.1	46.7	34.4
		MCDB	34.3	39.9	52.3	62.9	72.8	80.5	82.5	80.8	77.2	66.7	52.3	37.8
	5%	WB	29.6	32.8	41.1	49.3	59.7	67.5	70.8	69.0	65.6	54.1	42.5	32.1
		MCDB	31.2	35.9	46.9	61.4	68.9	76.3	80.0	77.1	73.1	61.7	47.6	34.3
	10%	WB	26.3	29.8	37.4	46.0	56.9	65.1	68.7	67.1	62.9	50.3	38.8	29.5
		MCDB	28.0	32.2	42.0	56.2	66.3	73.2	77.6	74.9	69.7	56.7	43.0	31.1
(35) Mean Daily Temperature Range	5% DB	MDBR	18.1	19.6	19.1	20.4	21.6	19.9	20.6	20.7	20.0	16.4	14.5	14.8
		MCDBR	17.5	20.6	24.4	29.3	29.4	24.3	23.6	23.9	25.0	25.1	22.0	17.0
		MCWBR	15.2	16.3	16.2	16.8	15.1	12.4	11.7	11.6	13.6	14.7	14.6	13.8
	5% WB	MCDBR	16.7	19.0	22.7	26.6	23.7	20.4	21.1	20.5	20.7	22.5	19.8	15.8
		MCWBR	14.8	15.2	15.8	16.1	13.9	12.2	12.5	11.4	12.9	14.3	14.1	13.2
(40) Clear-Sky Solar Irradiance	taub		0.266	0.300	0.301	0.348	0.367	0.374	0.378	0.378	0.351	0.306	0.279	0.262
	taud		2.319	2.214	2.391	2.360	2.367	2.373	2.381	2.401	2.455	2.566	2.516	2.390
	Ebn at Noon		260	271	293	287	285	282	279	274	272	272	259	250
	Edh at Noon		23	32	31	36	37	37	36	34	30	23	19	19
(44) All-Sky Solar Radiation	RadAvg		495	862	1220	1527	1695	1839	1966	1655	1213	740	442	350
	RadStd		51	79	126	139	104	131	94	126	89	116	45	41

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page